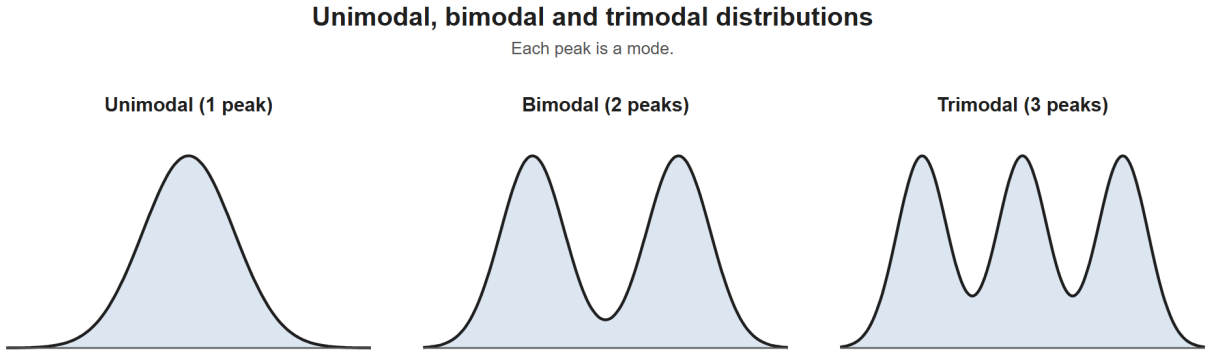
# Lesson 4.5 - Skewness with Box Plots

> **Video:** Skewness - Right, Left & Symmetric Distribution - Mean, Median, & Mode With Boxplots by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

## Number of modes

The **mode** is the value with the highest frequency, which is the **peak** of the distribution.

| Shape | Peaks | Name |
|---|---|---|
| One peak | 1 | **Unimodal** |
| Two peaks | 2 | **Bimodal** |
| Three peaks | 3 | **Trimodal** (or, in general, **multimodal**) |




The number of peaks gives the number of modes.

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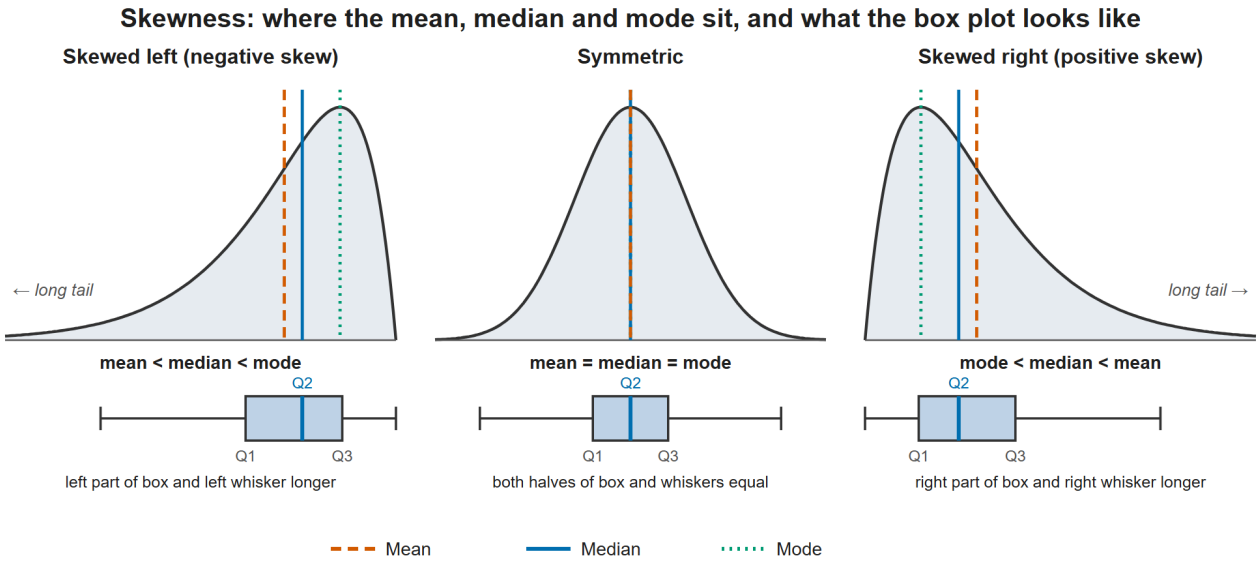
## The three shapes (unimodal distributions)

| Shape | Tail | Order of mean, median, mode |
|---|---|---|
| **Symmetric** | Left and right sides mirror each other | mean = median = mode |
| **Skewed right** (positive skew) | Long tail to the **right** | mode < median < **mean** |
| **Skewed left** (negative skew) | Long tail to the **left** | **mean** < median < mode |

Reading left to right along the x-axis:

- **Skewed right:** peak (mode) → median → mean → long tail on the right
- **Skewed left:** long tail on the left → mean → median → peak (mode)

> **Memory aid:** the **mean is pulled toward the tail**. The mode stays at the peak, and the median sits between them.




Curves: Gamma(2, 1), a normal curve, and the mirror image of Gamma(2, 1). Whiskers drawn to the 1.5 × IQR fences.

Where the mean, median and mode fall for each shape, with the matching box plot underneath. The mean is always pulled toward the tail.

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## Reading skewness from a box plot

Let the left part of the box be $Q_2 - Q_1$ and the right part be $Q_3 - Q_2$.

### Symmetric

```
|-------[=====|=====]-------|
```

- $Q_3 - Q_2 = Q_2 - Q_1$ (equal halves of the box)
- Whiskers of equal length

### Skewed right (positive)

Any of these indicates right skew:

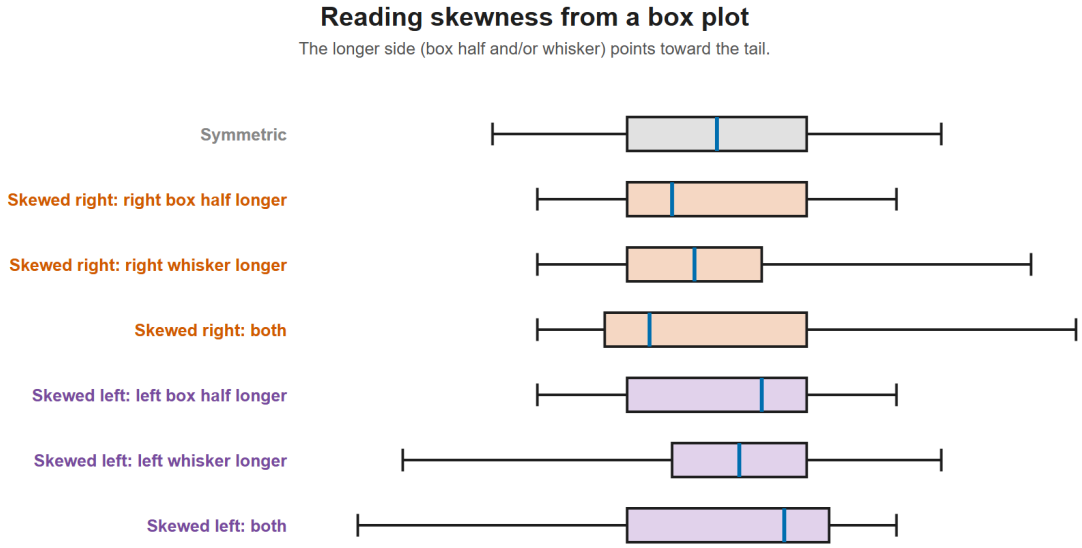
```
(a) right half of box longer:       |----[==|=======]----|
(b) equal box halves, longer right
    whisker:                        |--[====|====]----------|
(c) both:                           |--[==|========]------------|
```

- $Q_3 - Q_2 > Q_2 - Q_1$, and/or the right whisker is longer.

### Skewed left (negative)

```
(a) left half of box longer:        |----[=======|==]----|
(b) equal box halves, longer left
    whisker:                        |----------[====|====]--|
(c) both:                           |------------[========|==]--|
```

- $Q_2 - Q_1 > Q_3 - Q_2$, and/or the left whisker is longer.




The three ways a box plot can show skew: a longer box half, a longer whisker, or both. The longer side always points toward the tail.

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## Key takeaways

- Unimodal, bimodal, multimodal = one, two, many peaks.
- Symmetric: mean = median = mode.
- Right skew: mean > median > mode. Left skew: mean < median < mode.
- On a box plot, the **longer side** (box half and/or whisker) points in the direction of the skew.